



UNIVERSITY OF
KWAZULU-NATAL™
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HEALTH SYSTEMS AS COMPLEX ADAPTIVE SYSTEMS

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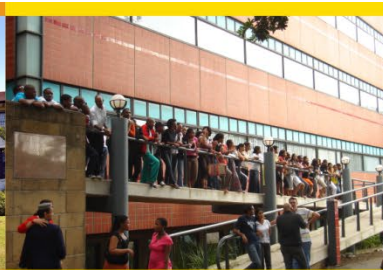
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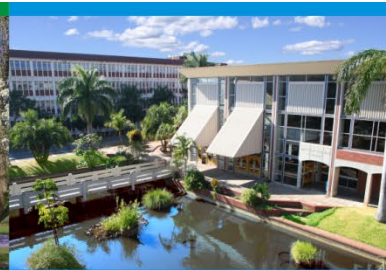
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OBJECTIVE OF THE LECTURE

- Cover some basic definitions
- Discuss the elements of a health system and interrelationships
- Explore the WHO Building blocks
- Look at the health system through a **DATA LENS**

WHAT STUDENTS SHOULD BE ABLE TO DO?

- By the end of the session students should move from naming components to reasoning about systems behaviour

1. **Describe** the six WHO health-systems building blocks and what each contributes

2. **Connect** building blocks using dependency and feedback relationships

3. **Identify** adaptation, time delays and unintended consequences

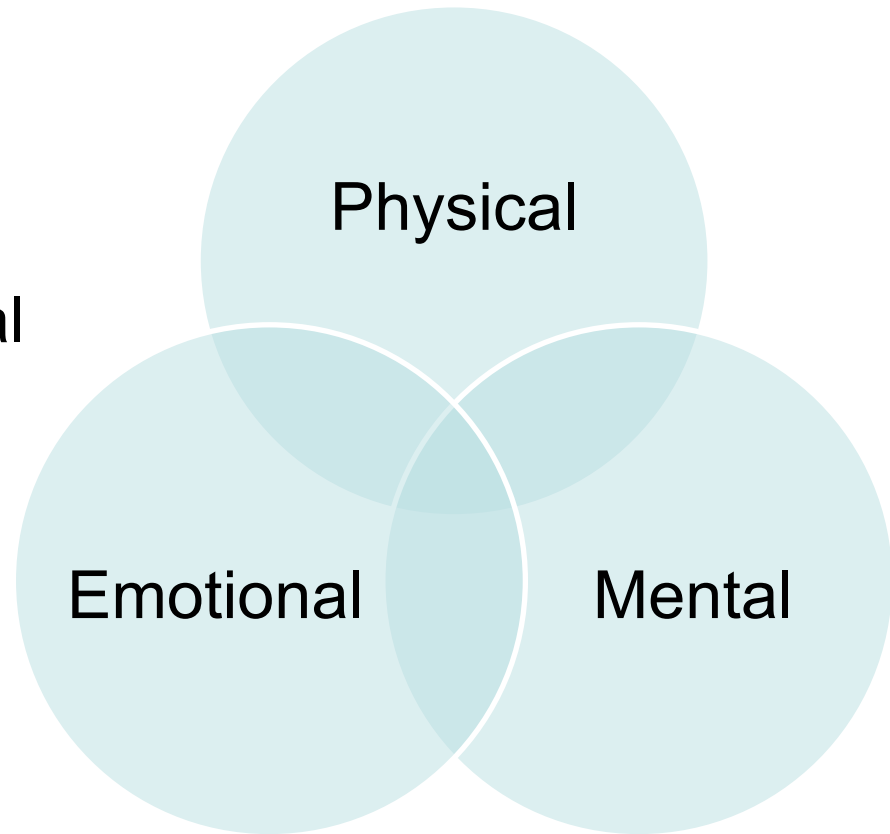
4. **Apply** a data-science lens, prediction, intervention and system response

Core shift: Which block is weak?  “What happens when the blocks interact?”

WHAT IS HEALTH?

State of well-being

- Free of disease and symptoms
- Active
- Good spirits / emotional state



What does health mean to you?

DEFINITION OF HEALTH

Health is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity.

The enjoyment of the highest attainable standard of health is one of the fundamental rights of every human being without distinction of race, religion, political belief, economic or social condition.

<https://www.who.int/about/governance/constitution>

HEALTH SYSTEMS

- Traditionally: described in terms of capacity indicators and activities (e.g. number of hospital beds, physicians and nurses, government programs.)
- Roemer: described by five characteristics:
 - Productive resources,
 - Organization of programs,
 - Economic support mechanisms,
 - Management methods
 - and service delivery

Roemer- 5 characteristics

- Productive resources,
- Organization of programs,
- Economic support mechanisms,
- Management methods
- and service delivery

Health system

- Health system as a set of functional components.
 - Londono and Frenk: system consists of four functions: financing, delivery, modulation and articulation
 - Anne Mills conceptualized health systems as loose framework of actors and functions- financing, regulation, resource allocation, and service provision

Health Economist definition

- Macro-level: *overall dimensions* of the health sector: size, shape, function
- micro-level explores behavior and dynamics of individual firms and households
- A set of relationships in which the means (i.e. structural components) are causally connected to the ends (i.e. goals.)

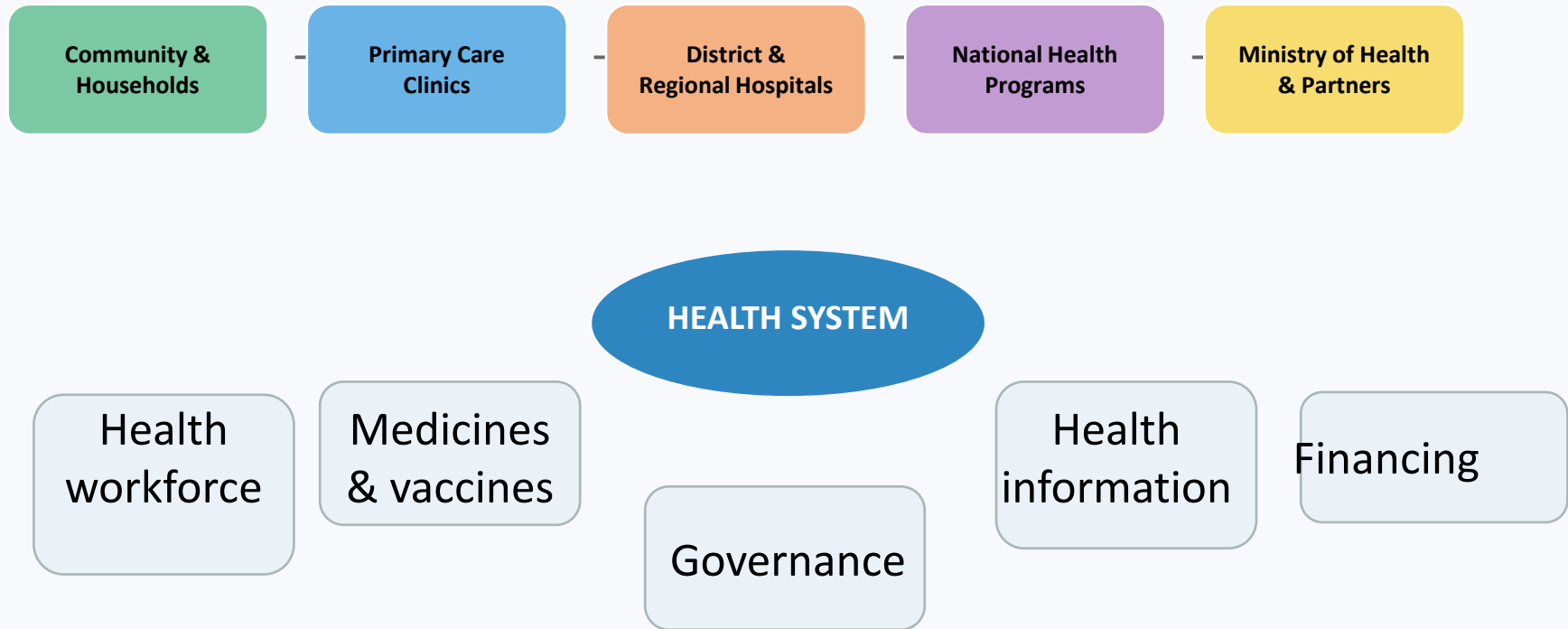
WHO- 2000

- A health system consists of all organizations, people and actions whose primary intent is to promote, restore or maintain health .
- This includes efforts to influence determinants of health as well as more direct health-improving activities

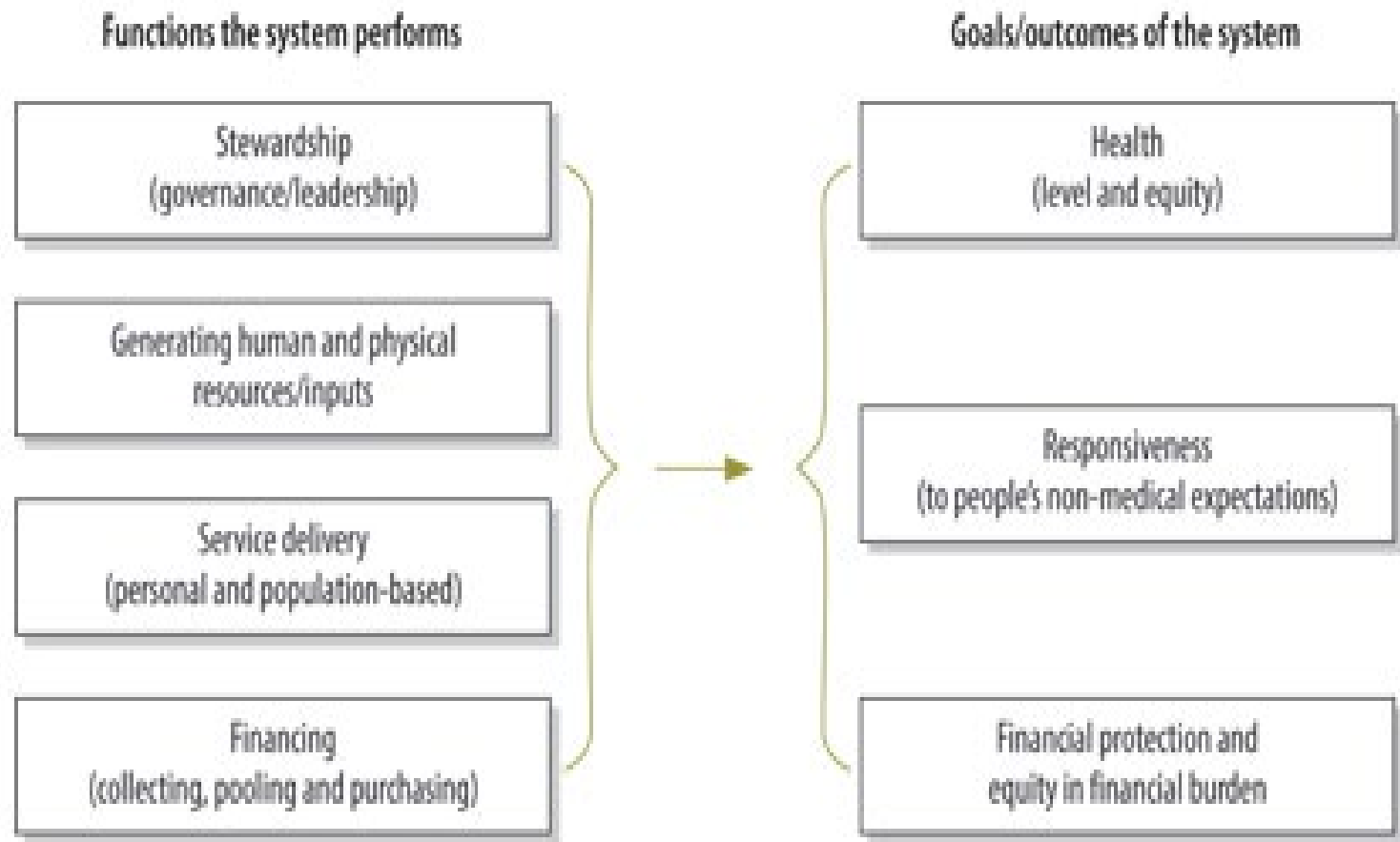


Typical Health System in a Low- and Middle-Income Country (LMIC)

Patients, referrals and information flow



Health system Goals



Health System Goals- Improved Health

- Health of individuals throughout the life course and include both premature mortality and non-fatal health outcomes
- Average level of population health and with the distribution of health within the population, namely health inequalities.

Responsiveness

- To enhance the responsiveness of the health system to the legitimate expectations of the population for the non-health improving dimensions of their interaction with the health system

Responsiveness

- Respect of persons and client orientation.
 - Respect for the dignity of the person
 - Respect for the autonomy of the individual to make choices about his/her own health.
 - Respect for confidentiality
 - Prompt attention to health needs.
 - Basic amenities
 - Access to social support networks for individuals receiving care.
 - Choice of institution and individual providing care

Fair financing and financial risk protection

- Households should not become impoverished or pay an excessive share of their income in obtaining needed health care
- Poor households should pay less towards the health system than rich households.
- Efficiency or composite goal performance.
- Efficiency is how well we achieve the socially desired mix of the five components of the three goals compared to the available resources

Leadership and Governance

- Ensuring the existence of strategic policy frameworks combined with:
 - ▣ Effective oversight
 - ▣ Coalition-building
 - ▣ Provision of appropriate regulations and incentives
 - ▣ Attention to system design
 - ▣ Accountability

Health governance (stewardship) context:

- ▣ Wide range of functions carried out by governments to:
 - Improve population health while ensuring:
 - ▣ Access to services
 - ▣ Quality of services
 - ▣ Patients' rights
- ▣ Examples:
 - Administrative details
 - Logistics and Operations
 - Planning and Policy Making
 - Monitoring and Evaluation

Healthcare Financing

Health/Health Care Financing

□ Good Health Financing:

▣ Two Main Characteristics:

- Raises adequate funds for health to ensure that people get to use needed services
- People who use health services are shielded from financial catastrophe or impoverishment associated with having to pay for them

▣ Health and Health Care are major political and economic issues

▣ Health financing impacts the analysis of:

- Health policies
- Fund sources
- Effectiveness and efficiency of health services for populations

▣ Health Financing Goals:

- ▣ Raising sufficient funds for health
- ▣ Ensure adequate spending on health
- ▣ Effective allocation of finite financial resources to different types of public and personal health services
- ▣ Pooling financial resources across population groups and sharing financial risks
- ▣ Using funds for health efficiently and equitably

Human Resources for Health

- To achieve the best health outcomes possible:
 - ▣ Sufficient numbers
 - ▣ Right mix of staff
 - ▣ System-wide deployment and distribution (equitable)
 - ▣ Established job-related norms
 - ▣ Enabling work environments
 - ▣ Just compensation/payment systems – right kind of incentives

Population needs determine the development and sustaining of health workforce

- ▣ Education, training and continuing competence
- ▣ Utilization, management and retention
- ▣ Strategic response to evolving and unmet health service needs
- ▣ Governance, leadership and partnerships for sustained HRH contributions to improved population outcomes
 - Regulation
 - Deployment
 - Compensation
 - Continuing career enhancement and development

Access to medicines and technology

- Ensured equitable access to:
 - ▣ Essential medical products
 - ▣ Effective, Safe, Cheap Medicines
 - ▣ Vaccines
 - ▣ Affordable and readily available Medical Technologies
- Governing Principles of:
 - ▣ Quality
 - ▣ Safety
 - ▣ Efficacy
 - ▣ Cost-effectiveness
 - ▣ Scientifically sound

Essential medicines and technologies save lives, reduce suffering and improve health

- ▣ Available
- ▣ Affordable
- ▣ Assured quality (evidence-based, GMPs, etc)
- ▣ Used properly by both provider and patient
- ▣ Price can be afforded by individuals and the system

Equity in access is crucial

Health Information and Research



- Health information is a national asset and used by many:
 - ▣ Policy-makers
 - ▣ Planners
 - ▣ Health care providers
 - ▣ Development partners
 - ▣ The general public
- Uses:
 - ▣ Track health system performance
 - ▣ Support better health policies
 - ▣ Make effective health-related decisions

Health Service Delivery

- Good health services:
 - ▣ Deliver effective, safe and quality health interventions to those who need them; when and where needed, with minimum waste of resources



Health Service Delivery

- Health services – most visible functions of any health system
 - Service provision: the way inputs (money, staff, equipment, drugs, etc.) are combined for the delivery of health interventions
 - Ensured availability of key resources as well as good service management and organization result in:
 - Improved coverage
 - Better quality of health services
- Ultimate aim:
 - Equity in health outcomes



Health Service Delivery



□ Key Elements:

- Organizing health services as networks of primary care backed up by hospitals and specialized care
- Providing a package of health benefits with clinical and public health interventions
- Ensuring access and quality of services
- Holding providers accountable for access and quality and ensuring consumer voice

Strengthening vs. supporting health systems

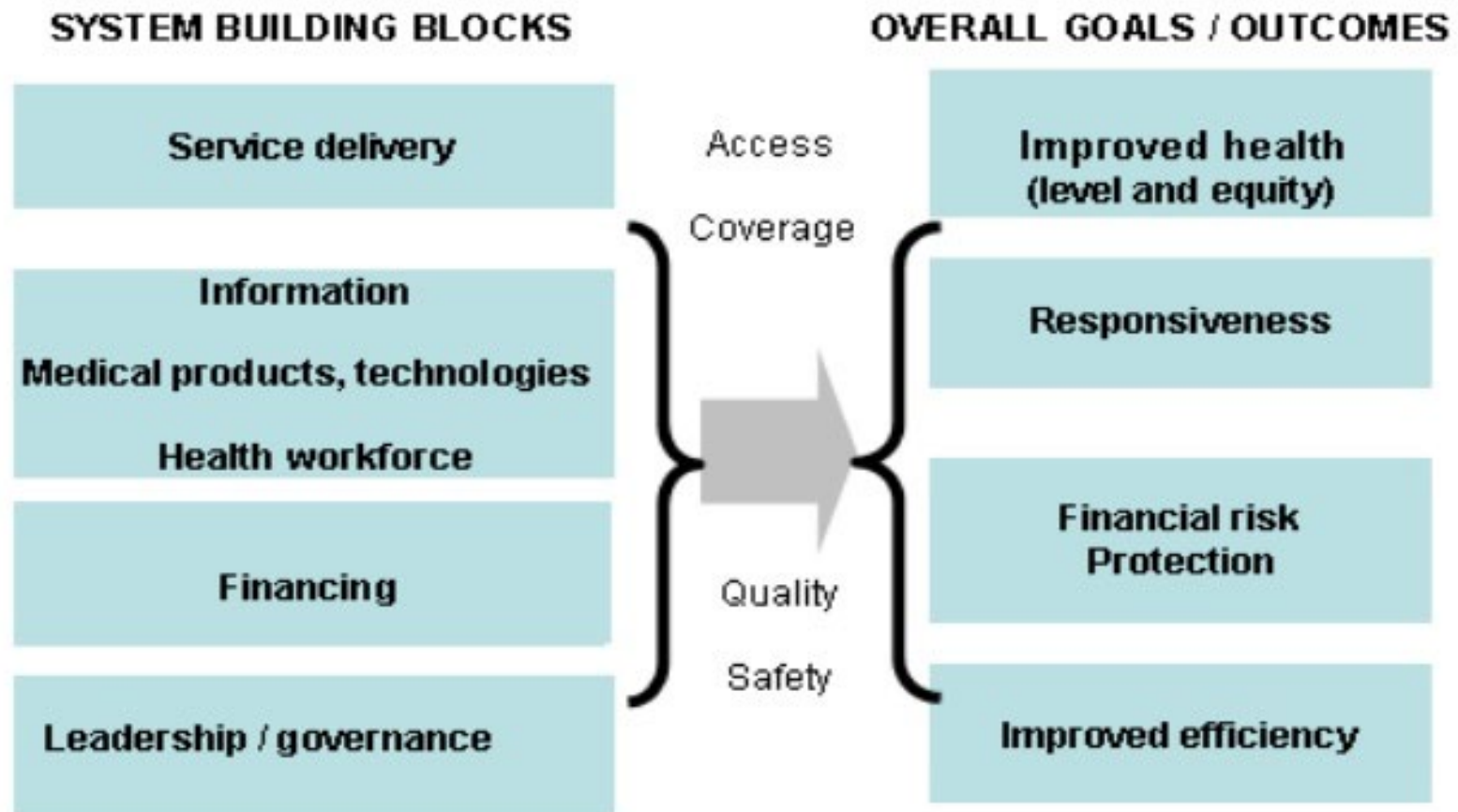
Parameter	Health System <u>Support</u>	Health System <u>Strengthening</u>
Scope	May be focused on a single disease or intervention	Activities have impact across health services and outcomes
Longevity	Effects limited to period of activity	Effects will continue after activities end
Approach	Provide inputs to address identified system gaps	Revise policies and institutional relationships to change behaviors and resource use to address identified constraints

Vertical, Horizontal and Diagonal approaches

- Vertical programs: focus on delivery of intervention targeting specific disease or condition (eg immunisation)
- Horizontal program: focus on health system components with relatively less attention on delivery outcomes
- Diagonal approach: Addressing health systems bottlenecks to specific vertical programs while achieving system-wide benefits

WHO Everybody's business 2009

Building blocks combine to meet health system goals



Opening challenge: change one thing

4 min

Think-pair-share (60–90 seconds).

A district increases clinic staffing by 20% to reduce waiting times.

1

**What might
improve?**

2

**What else might
change?**

3

**What could
become worse?**

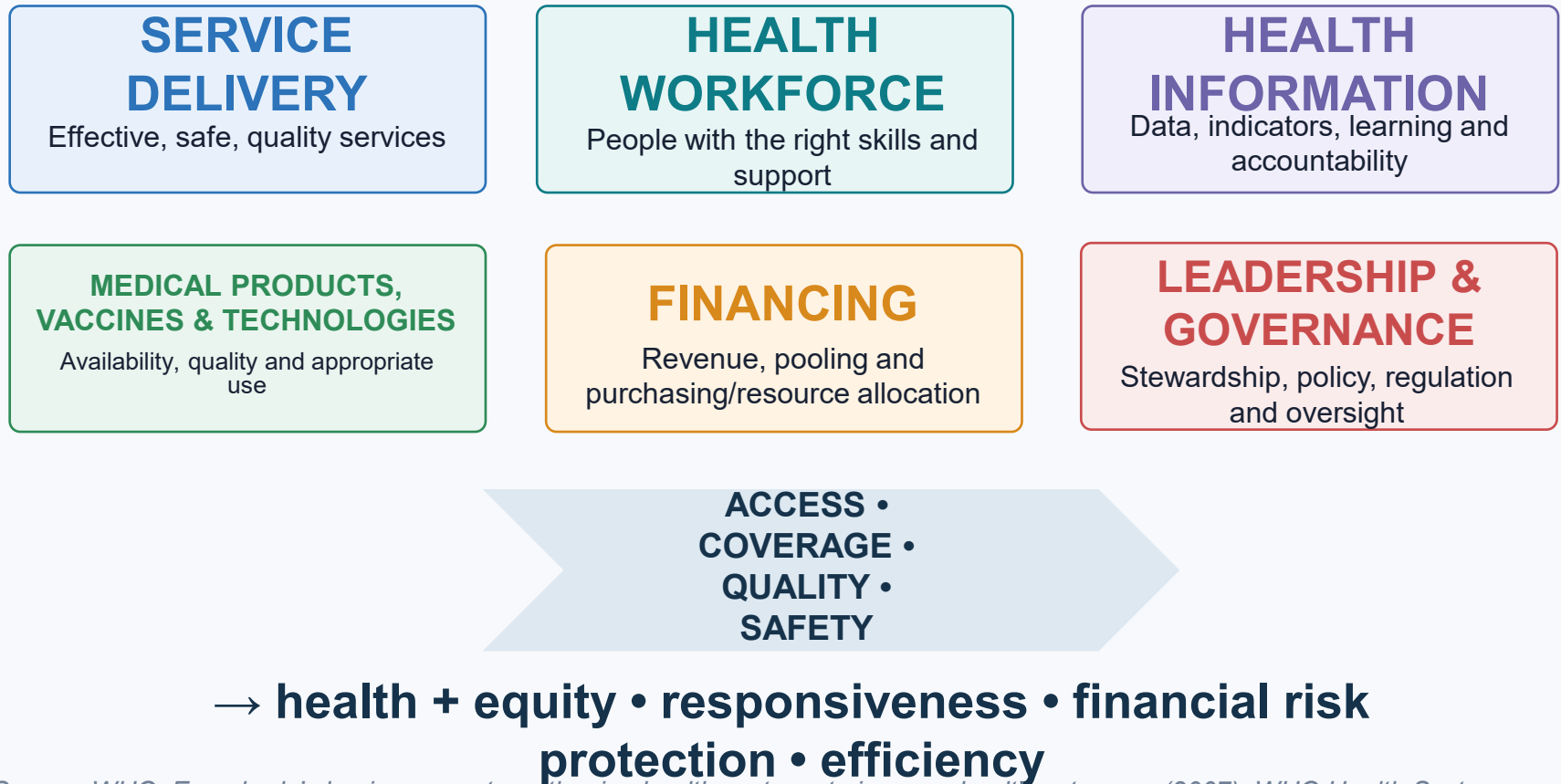
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**What new data
would you need?**

Debrief prompt: “Why is staffing not an isolated variable?”

The WHO health-system framework

Six core components used to analyse and strengthen health systems.



Source: WHO, *Everybody's business — strengthening health systems to improve health outcomes* (2007); WHO Health System Building Blocks (current WHO National Health Planning Tools).

The six blocks — what a data scientist sees?

Use the blocks as domains of data, decisions and dependencies.

1. Service delivery: visits, waiting time, quality, outcomes

2. Health workforce: staffing, skills, workload, vacancies, absenteeism

3. Health information: registries, routine data, surveillance, dashboards, interoperability

4. Products and technology: stock-outs, procurement, equipment, diagnostics, medicines

5. Financing Budgets, expenditure, incentives, purchasing, access barriers

6. Leadership & Governance: targets, rules, priorities, accountability, coordination

Important: the same outcome can be influenced by several blocks at once.

1 • Service delivery

What is delivered, to whom, when, where and with what quality?

1. Core Concept

- Effective, safe, quality health services that are accessible to those who need them.
- Includes availability, accessibility, continuity, comprehensiveness, quality and people-centredness.
- The block is concerned with the organisation and provision of care—not only the number of services delivered.

2. Key dimensions

- Availability/ access
- Quality/safety
- Equity/ responsiveness

6

The building block describes a domain of the health system; it does not operate in isolation.

Source: WHO. Everybody's business — strengthening health systems to improve health outcomes. Geneva: WHO; 2007.

1 • Service delivery

How this block connects resources, actions, behaviours and outcomes?

Systems Role

- Converts system resources into services and patient experiences.
- Links population need with service capacity, referral pathways and patterns of utilisation.
- Performance depends on coordination with workforce, medicines, financing and information—not on service delivery alone.

Data Science bridge

- Utilisation, waiting time, coverage, referral flows, quality, outcomes and geographic access.

2 • Health workforce

People with the right skills, in the right place, at the right time, with appropriate support.

1. Core Concept

- Encompasses numbers, distribution, skill mix, competencies, productivity, motivation and working conditions.
- Workforce capacity is therefore more than headcount: two facilities with the same staffing complement may have very different effective capacity.
- The workforce is both a resource and an active human component of the system.

Key Dimensions

- Availability / access
- Quality / safety
- Capacity / distribution

8

The building block describes a domain of the health system; it does not operate in isolation.

Source: WHO. Everybody's business — strengthening health systems to improve health outcomes. Geneva: WHO; 2007.

2 • Health workforce

How this block connects resources, actions, behaviours and outcomes

Systems role

- Acts as the main mechanism through which policy, financing, technology and information become care.
- Workload, incentives, supervision and working conditions influence performance and retention.
- Changes in one part of the system can alter workforce behaviour and therefore change system performance.

Data Science bridge

- Staffing, skill mix, workload, absenteeism, turnover, productivity, deployment and time use.

3 • Health information systems

Information is generated, organised and used to monitor, learn, coordinate and make decisions.

1. Core concept

- Includes data collection, reporting systems, records, indicators, analysis, dissemination and use.
- A functioning information system requires not only data production but also data quality, interpretation and timely use.
- Information is therefore both an input to the health system and an output of system activity.

2. Key Dimension

- Availability / access
- Quality / safety
- Data / information flow

The building block describes a domain of the health system; it does not operate in isolation.

Source: WHO. Everybody's business — strengthening health systems to improve health outcomes. Geneva: WHO; 2007.

3 • Health information systems

How this block connects resources, actions, behaviours and outcomes

System role

- Provides the system with memory and feedback: what happened, where, for whom and with what results?
- Information supports accountability, coordination, learning and adaptation.
- The act of measuring can itself influence behaviour—especially when indicators become targets.

Data Science bridge

- Data quality, completeness, timeliness, linkage, interoperability, reporting burden, dashboards and decision use.

Source: WHO. Everybody's business; WHO. Monitoring the Building Blocks of Health Systems.

1
1

4 • Medical products, vaccines & technologies

People need reliable access to safe, effective, quality-assured and appropriately used products and technologies.

1. Core concept

- Covers medicines, vaccines, diagnostics, devices and other health technologies.
- Theoretical concerns include availability, affordability, quality, procurement, supply and rational use.
- Access depends on the whole chain—from selection and forecasting to procurement, distribution and prescribing.

2. Key dimensions

- Availability / access
- Quality / safety
- Supply / quality

The building block describes a domain of the health system; it does not operate in isolation.

Source: WHO. Everybody's business — strengthening health systems to improve health outcomes. Geneva: WHO; 2007.

1
2

4 • Medical products, vaccines & technologies

How this block connects resources, actions, behaviours and outcomes

Systems role

- Operates as a coupled supply-and-demand system rather than a simple stockholding function.
- Consumption patterns, prescribing behaviour, procurement delays and stock levels influence one another.
- Shortages can shift behaviour elsewhere in the system, creating substitution, repeat visits or changes in utilisation.

Data Science bridge

- Stock levels, stock-outs, lead times, consumption, procurement performance, prescribing and geographic distribution.

1
3

Source: WHO. Everybody's business; WHO. Monitoring the Building Blocks of Health Systems.

5 • Health financing

Resources are mobilised, pooled and allocated so that people can access needed services without undue financial hardship.

1. Core concept

- Three core functions are commonly distinguished: revenue raising, pooling and purchasing/resource allocation.
- Financing affects both availability of services and the incentives faced by providers and institutions.
- Theoretical performance concerns include efficiency, equity, sustainability and financial protection.

2. Key dimensions

- Availability / access
- Quality / safety
- Equity / efficiency

The building block describes a domain of the health system; it does not operate in isolation.

Source: WHO. Everybody's business — strengthening health systems to improve health outcomes. Geneva: WHO; 2007.

5 • Health financing

How this block connects resources, actions, behaviours and outcomes

Systems role

- Financing is not only a quantity of money; it is also a set of rules that shapes behaviour and priorities.
- How funds are allocated changes service mix, staffing, purchasing and access.
- Payment and allocation mechanisms can produce intended effects and also create strategic responses by actors.

Data Science bridge

- Budgets, expenditure, claims, funding flows, unit costs, resource allocation, utilisation and financial protection.

Source: WHO. Everybody's business; WHO. Monitoring the Building Blocks of Health Systems.

6 • Leadership & governance

Stewardship: setting direction, establishing rules, coordinating action and ensuring accountability.

1. Core concept

- Includes policy, regulation, strategic direction, oversight, participation, transparency and institutional accountability.
- Governance determines who has authority, which goals are prioritised and how competing interests are managed.
- Leadership is therefore a relational and institutional function, not simply an individual attribute.

2. Key dimensions

- Availability / access
- Quality / safety
- Authority / accountability

The building block describes a domain of the health system; it does not operate in isolation.

Source: WHO. Everybody's business — strengthening health systems to improve health outcomes. Geneva: WHO; 2007.

6 • Leadership & governance

How this block connects resources, actions, behaviours and outcomes

Systems role

- Creates the rules and decision structures within which the other building blocks operate.
- Governance influences coordination, incentives, information use and the distribution of resources and authority.
- Because actors respond to rules and signals, governance choices can generate emergent and sometimes unintended system behaviour.

Data Science bridge

- Decision rights, policies, accountability, performance management, regulation, coordination, participation and implementation.

1
7

Source: WHO. Everybody's business; WHO. Monitoring the Building Blocks of Health Systems.

1 • Service delivery

What is delivered, to whom, when, where and with what quality?

SD What it means

Effective, safe, quality services that are accessible when and where needed, with minimum waste.

DS Data scientist lens

Utilisation, waiting time, throughput, referral flows, quality indicators, outcomes, geographic access.

↔ Interaction

Service volume ↔ workforce capacity ↔ medicines/equipment ↔ financing ↔ patient behaviour.

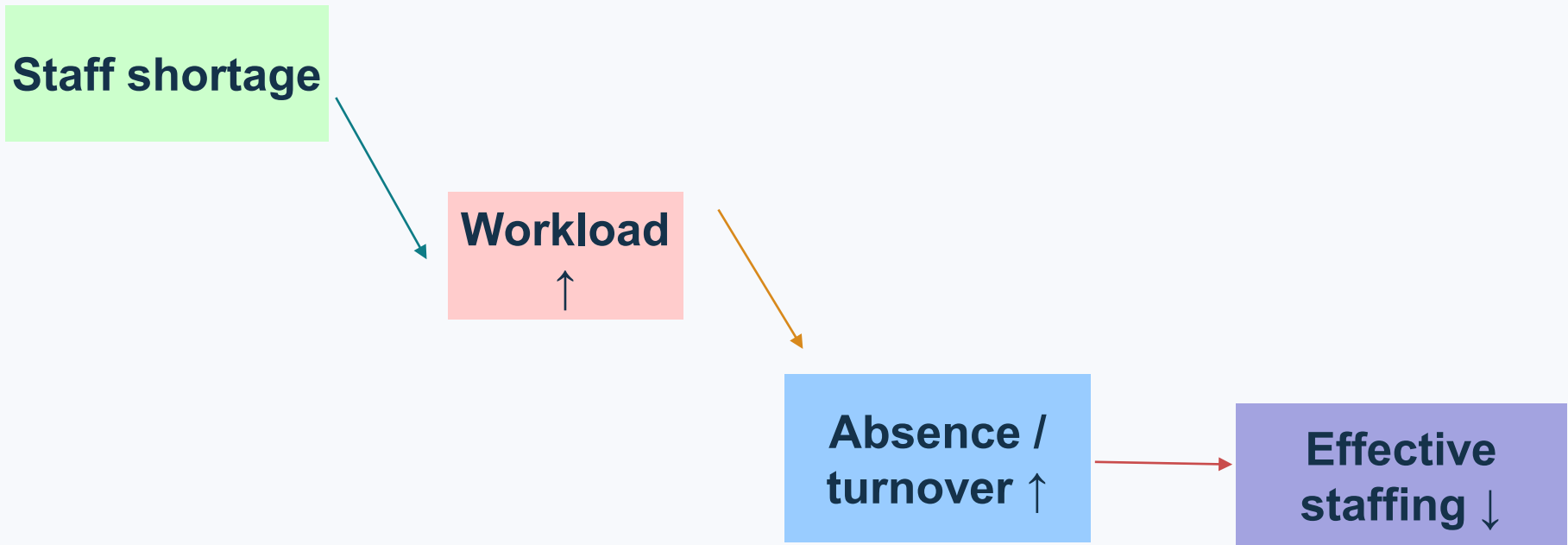
A waiting-time model predicts that Mondays are busiest. What happens when management changes staffing on Mondays?

The intervention can change the data-generating process.

WHO framework: service delivery is one of the six core health-system components.

2 • Health workforce

Capacity is not just headcount — it is skills, deployment, workload, motivation and time.



Are “vacancies” and “effective capacity” the same thing? What time lag might exist between workload and turnover?

3 • Health information systems

Information is both an input to the system and an output of the system.

1 Measure

What is recorded?

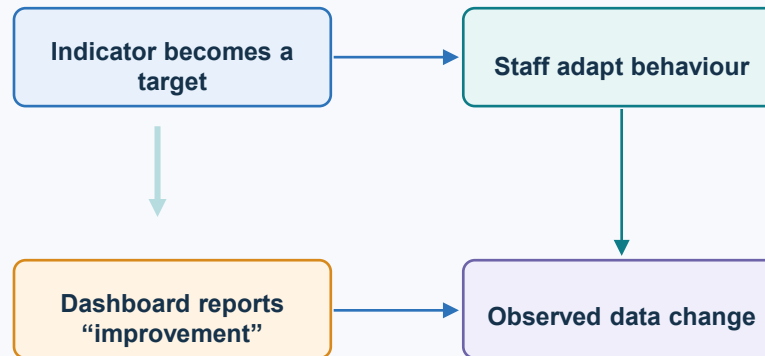
2 Interpret

What does it mean?

3 Act

What decision follows?

But measurement changes behaviour

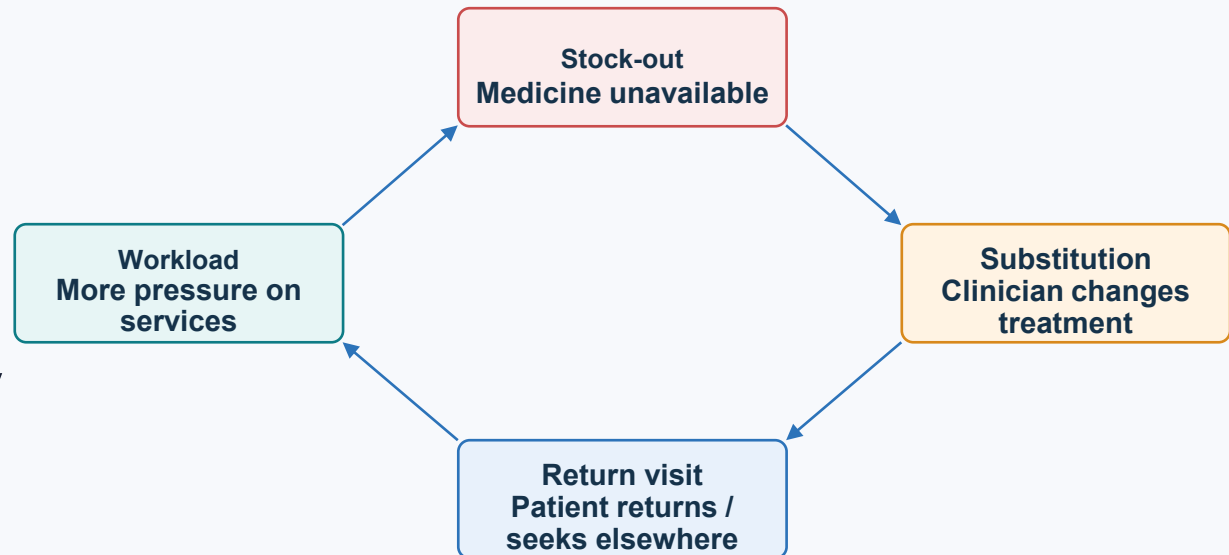


What happens to an indicator when people know they are being judged by it

a useful measure can become a poor proxy when optimised directly.

4 • Medical products, vaccines & technologies

Availability, quality and appropriate use interact with every other block.

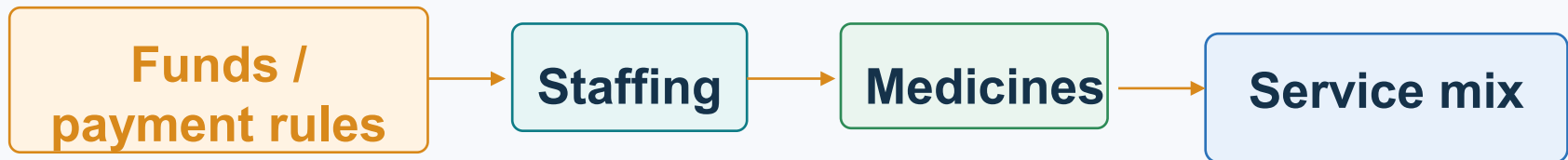


- stock-out prediction
- supply-chain anomaly detection
- demand forecasting
- substitution patterns
- spatial access / facility networks

Key idea: local coping strategies can shift pressure elsewhere.

5 • Health financing

Financing changes incentives, capacity and access — not just “how much money”.



But incentives can create adaptation

Payment metric → behaviour → activity recorded → future allocation

“What would you monitor to distinguish a real improvement from a change in coding / case-mix / service mix?”

6 • Leadership & governance

Who sets rules, allocates attention, interprets evidence and coordinates the system?

Policy

What should happen?

Targets

What gets measured?

Accountability

Who responds?

Coordination

Who connects the pieces?

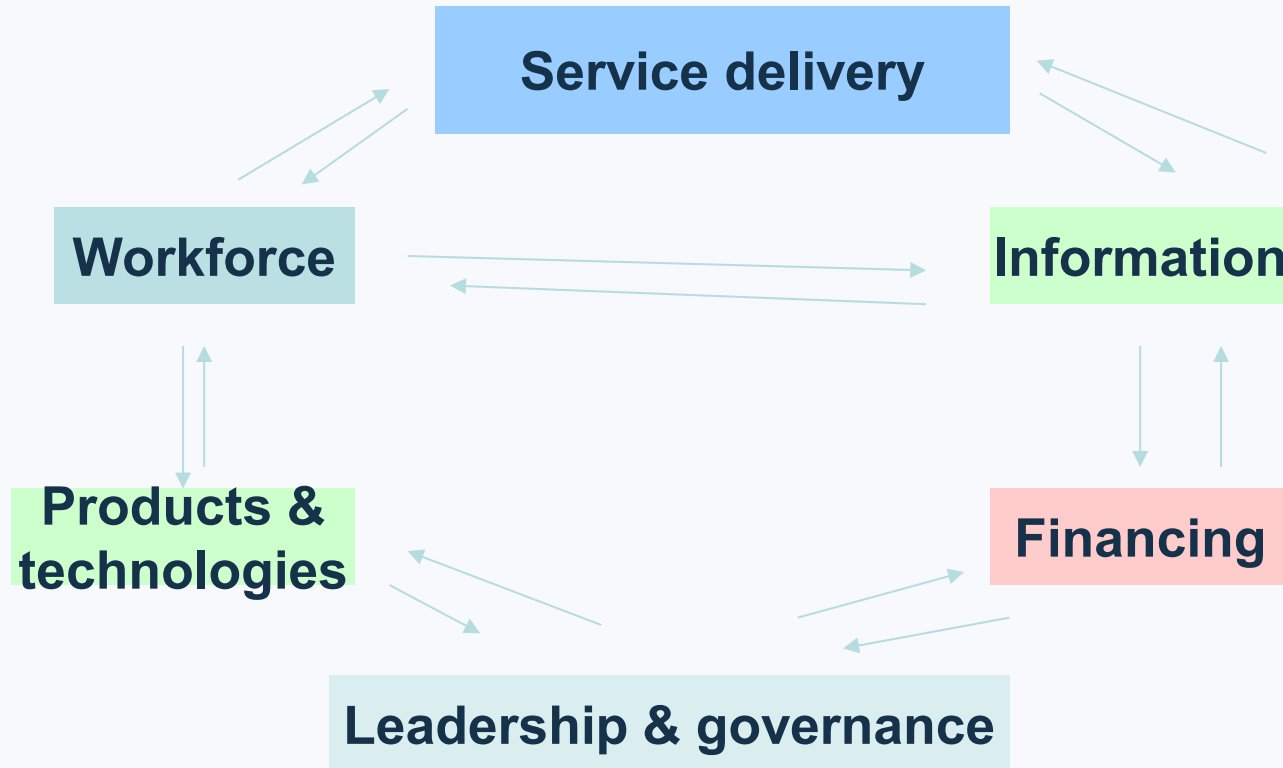
Governance shapes how every other block behaves

“Who has the power to act on this insight — and what incentives or constraints will shape their response?”

EVIDENCE → DECISION → RESPONSE → NEW EVIDENCE

From six blocks to one connected system

The building blocks are analytically separable — operationally they are entangled.



Example relationship: governance → financing → workforce → service delivery → information → governance

The complexity lens: four properties to look for

These are the “extra questions” we add to the WHO building blocks.

1

Relationships

Blocks affect one another;
causes are rarely one-to-one.

2

Feedback

Changes feed back into future
demand, capacity or behaviour.

3

Adaptation

Patients, clinicians, managers
and organisations respond to
interventions.

4

Unintended effects

Local optimisation can
create effects elsewhere in
the system.

**Complex ≠ merely complicated: the key is interaction +
adaptation over time.**

A data scientist is inside the system, not outside it

The strongest analytics workflow asks what happens after the model is used.



Critical question: “What will people do because the model exists?”

This is where prediction meets causal inference, fairness, model monitoring and systems thinking.

Group case: reducing outpatient waiting time

Use the WHO building blocks + complexity lens. Groups of 3–5.

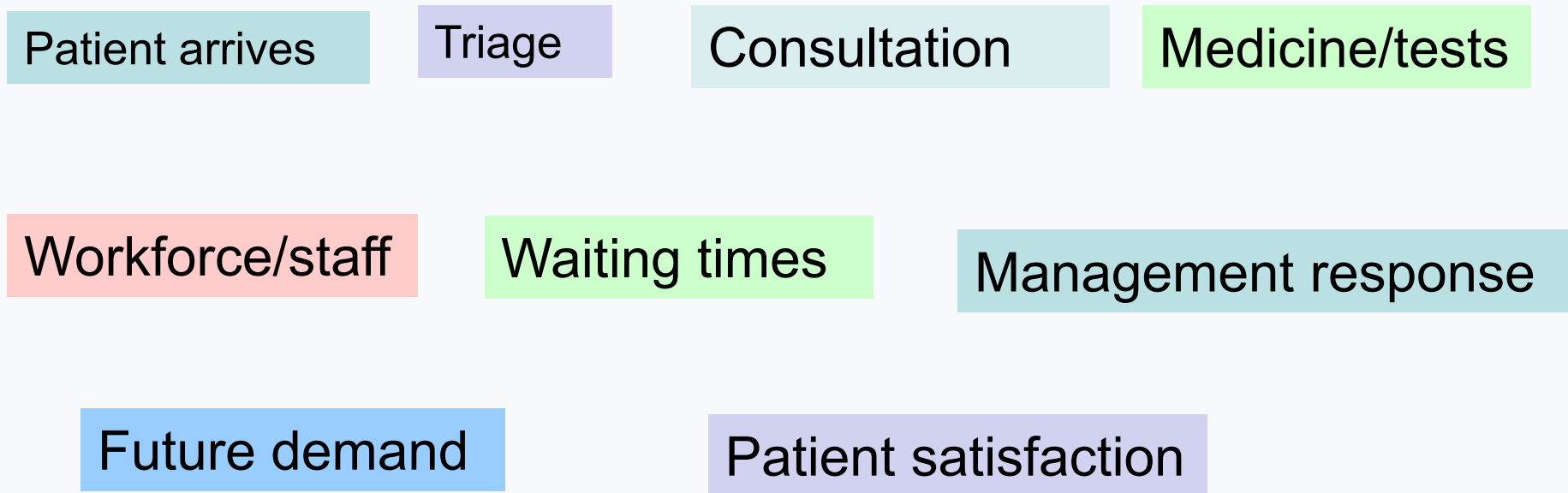
A district clinic has rising waiting times. Leadership asks the data team to “find the bottleneck and build a prediction model.”

- A. Map the blocks** Which WHO blocks could influence waiting time?
- B. Map the data** What variables would you collect?
- C. Map the feedback** How could an intervention change the future data?
- D. Find unintended effects** What might improve while another part worsens?

Deliverable (last 2 minutes): draw one causal loop and propose one data source you would monitor after the intervention.

Debrief: the “bottleneck” is often a network

There may be several plausible intervention points — with different downstream effects.



Which arrow would you investigate first?

Which relationship could be nonlinear?

Where is the likely time delay?

What would surprise you in the data?

Take-home: keep the blocks — change the question

The WHO framework is the map of components. Complexity asks us to trace the movement between them.

FROM “Which block is weak?”

TO “How do blocks interact?”

FROM “What predicts the outcome?”

TO “What changes when we intervene?”

FROM “Is the model accurate?”

TO “How will the system respond to the model?”

For data scientists: model the system that produces the data — and the response your model creates

Core references and further reading

Use these sources for the conceptual framework and for extending the lecture.

1. World Health Organization. Everybody's business — strengthening health systems to improve health outcomes: WHO's framework for action. Geneva: WHO; 2007.
2. World Health Organization. Monitoring the Building Blocks of Health Systems: A Handbook of Indicators and their Measurement Strategies. Geneva: WHO; 2010/2017 edition.
3. World Health Organization. Health system building blocks. National Health Planning Tools. Current WHO resource.
4. Alliance for Health Policy and Systems Research. Systems thinking for health systems strengthening. Geneva: WHO; 2009.
5. World Health Organization. Systems thinking for health systems strengthening — “from bricks to relationships” framing. WHO AHPSR, 2026.

Primary framework: WHO six building blocks • Teaching lens: complex adaptive systems + data science

Web sources accessed for slide preparation: WHO Health System Building Blocks; WHO Everybody's business (2007); WHO Monitoring the Building Blocks; WHO AHPSR systems-thinking material (2026).